# Enterprise Synthesis Team

# PHARMACY MANAGEMENT SYSTEM

# PROJECT CHARTER

A four-month Agile / Incremental software development project

| | |
|---|---|
| **PROJECT LEAD** | Bahram Jafaree |
| **DEVELOPMENT TEAM** | Said Yousaf Hashimi • Ali Seena • Alireza Alizada • Zaki Ahmadi |
| **DURATION** | 4 Months |
| **STATUS** | PROPOSED |

*Prepared for academic project planning, development, review and final demonstration*

# 01 Executive Snapshot

A one-page decision view of the project

| 04 | 05 | 08 | 01 |
|---|---|---|---|
| MONTHS | CORE TEAM | DELIVERY PHASES | INTEGRATED PMS |

**PROJECT INTENT**

Create a secure, reliable and user-friendly pharmacy management platform that centralizes medicines, inventory, purchases, sales, suppliers and reporting—reducing manual work while improving operational accuracy.

**PRIMARY CONSTRAINT**

The approved initial scope must be delivered within four months. Any major change must be evaluated for scope, schedule, quality and resource impact before acceptance.

# 02 Project Identification

| Item | Details |
|---|---|
| **Project Title** | Pharmacy Management System (PMS) |
| **Project Type** | Software Development Project |
| **Project Duration** | 4 Months |
| **Project Manager / Team Leader** | Bahram Jafaree |
| **Development Team** | Said Yousaf Hashimi, Ali Seena, Alireza Alizada, Zaki Ahmadi |
| **Project Methodology** | Agile / Incremental Development |
| **Target Users** | Pharmacy owners, pharmacists, managers, cashiers and inventory staff |
| **Project Status** | Proposed |

# 03 Project Purpose

The purpose of the Pharmacy Management System (PMS) is to develop a computerized system that helps pharmacies efficiently manage their daily operations.

The system will replace or reduce manual processes such as recording medicines, managing inventory, processing sales, maintaining supplier information, tracking purchases and preparing reports.

The PMS will provide pharmacy staff with an organized, secure and easy-to-use platform for managing pharmacy information and improving the accuracy and efficiency of daily operations.

# 04 Project Background

Many pharmacies still depend on manual registers, spreadsheets or disconnected systems to manage medicines, sales, purchases and inventory. These approaches can result in data-entry errors, difficulty tracking stock, expired medicines, inaccurate records and delays in preparing reports.

The proposed PMS will centralize important pharmacy operations into one system. It will allow authorized users to manage medicine information, monitor stock levels, record purchases and sales, manage suppliers, track expiration dates and generate useful reports.

The project will be developed over four months by the assigned software development team.

# 05 Project Objectives

1. Develop a computerized system for managing pharmacy operations.
2. Maintain accurate records of medicines and their quantities.
3. Manage medicine purchases and sales efficiently.
4. Monitor inventory and identify low-stock medicines.
5. Track medicine expiration dates.
6. Maintain supplier and customer-related records where required.
7. Provide secure login and user-management mechanisms.
8. Reduce errors caused by manual data entry and record keeping.
9. Generate useful management and financial reports.
10. Provide a simple and user-friendly interface.
11. Protect pharmacy data through authentication, authorization and appropriate database security.
12. Complete, test, document and demonstrate the system within the four-month schedule.

# 06 Project Scope

**The boundary of the initial four-month release**

## 6.1 In Scope

### A. User Authentication & Management

- Login and logout
- User account management
- Role-based access control
- Password protection
- Different permissions for administrators and staff

### B. Medicine Management

- Add, edit, delete/deactivate medicine records
- Search and filter
- Categories and manufacturers
- Prices and quantities
- Batch number and expiration date

### C. Inventory Management

- Current stock monitoring
- Stock additions/reductions
- Low-stock notifications
- Expiration monitoring
- Stock adjustment
- Inventory history

### D. Sales Management

- Create sales transactions
- Search medicines
- Calculate totals
- Record quantities sold
- Receipts/invoices
- Sales history

### E. Purchase Management

- Record purchases
- Purchase information
- Supplier records
- Automatic inventory updates

- Purchase history

### F. Supplier Management

- Add/edit suppliers
- Contact information
- Supplier purchase history

### G. Reporting

- Daily sales
- Weekly/monthly sales
- Purchases
- Inventory
- Low stock
- Expired/near-expiration medicines
- Medicine sales history

### H. Dashboard

- Total medicines
- Available stock
- Low-stock medicines
- Expired medicines
- Today's sales
- Recent transactions
- Purchase information

## 6.2 Out of Scope

- Online medicine delivery.
- Integration with national healthcare systems.
- Direct connection to banks or payment gateways.
- Advanced AI diagnosis or medical recommendations.
- Automatic prescription verification with external healthcare databases.
- Large-scale multi-branch pharmacy management.
- Advanced accounting/tax integration.
- Hardware integration such as barcode scanners or receipt printers, unless time permits.

**FUTURE-VERSION CANDIDATES**

Out-of-scope capabilities may be evaluated for later releases after the core PMS is stable, tested and approved.

# 07 Major Project Deliverables

| No. | Deliverable |
|---|---|
| 1 | Project Charter |
| 2 | Requirements Specification |
| 3 | Feasibility Study |
| 4 | System Analysis |
| 5 | Use Case Diagrams |
| 6 | Database Design / ER Diagram |
| 7 | System Architecture |
| 8 | UI/UX Design |
| 9 | Database Implementation |
| 10 | Pharmacy Management System |
| 11 | Testing Documentation |
| 12 | User Manual |
| 13 | Technical Documentation |
| 14 | Final Project Report |
| 15 | Final System Demonstration |

# 08 Stakeholders

| Stakeholder | Main Interest |
|---|---|
| **Pharmacy Owner** | Accurate management, reports and business control |

| Stakeholder | Main Interest |
|---|---|
| **Pharmacy Manager** | Inventory, purchases, sales and reports |
| **Pharmacist** | Quick access to medicine information and stock |
| **Cashier** | Fast and accurate sales processing |
| **Inventory Staff** | Stock monitoring and expiration tracking |
| **Supplier** | Accurate purchase records |
| **Project Supervisor** | Successful completion and documentation |
| **Development Team** | Successful development and learning experience |

# 09 Project Team

| Team Member | Role | Main Responsibilities |
|---|---|---|
| **Bahram Jafaree** | Team Leader / Project Manager | Planning, coordination, task assignment, progress monitoring, communication and final supervision |
| **Said Yousaf Hashimi** | System Analyst / Backend Developer | Requirements analysis, architecture, backend development and business logic |
| **Ali Seena** | Frontend / UI Developer | User interface, screens, forms, dashboard and frontend implementation |
| **Alireza Alizada** | Database Developer | Database design, ER diagram, implementation, relationships and data integrity |
| **Zaki Ahmadi** | QA & Documentation | Test planning, software testing, bug reporting, documentation and user manual |

Team members may collaborate across responsibilities when necessary.

# 10 Project Schedule — Four Months

| Phase | Activities | Timing |
|---|---|---|
| **Phase 1** | Project initiation, requirements gathering and feasibility study | Weeks 1–2 |
| **Phase 2** | System analysis, use cases and requirements specification | Weeks 3–4 |
| **Phase 3** | UI/UX design, architecture and database design | Weeks 5–6 |
| **Phase 4** | Database and core system development | Weeks 7–9 |
| **Phase 5** | Medicine, inventory, purchase and supplier modules | Weeks 10–12 |
| **Phase 6** | Sales, dashboard and reporting modules | Weeks 13–14 |
| **Phase 7** | Integration, testing and bug fixing | Weeks 15–16 |
| **Phase 8** | Final documentation, deployment and presentation | Final days / Week 16 |

## 10.1 Monthly Roadmap

| Month | Primary Focus | Key Outputs |
|---|---|---|
| **Month 1** | Planning & Analysis | Charter, requirements, feasibility, analysis, use cases, initial DB design |
| **Month 2** | Design & Core Development | UI/UX, architecture, database, authentication, medicine management |
| **Month 3** | Feature Development | Inventory, sales, purchases, suppliers, dashboard and reports |
| **Month 4** | Testing & Delivery | Integration, testing, bug fixing, UAT, documentation, deployment and presentation |

# 11 Constraints & Risks

## 11.1 Constraints

13. The project must be completed within four months.
14. The development team has limited time and resources.
15. Team members may have academic or other responsibilities.
16. The project may depend on the availability of real pharmacy requirements.
17. The system must be developed using available hardware and software resources.
18. The initial version should focus on essential pharmacy-management functions rather than unnecessary advanced features.
19. Changes to major requirements after development begins may affect the schedule.

## 11.2 Risk Register

| Risk | Impact | Mitigation |
|---|---|---|
| **Requirements change** | High | Confirm requirements before development |
| **Project delays** | High | Weekly progress monitoring |
| **Team member unavailable** | Medium / High | Share documentation and knowledge |
| **Database errors** | High | Regular testing and backups |
| **Security vulnerabilities** | High | Authentication, authorization and validation |
| **Data loss** | High | Regular database backups |
| **Scope creep** | High | Require approval for major new features |
| **Technical difficulties** | Medium | Research, testing and team collaboration |
| **Integration problems** | Medium | Integrate modules gradually |
| **Insufficient testing** | High | Maintain a formal testing plan |

# 12 Project Assumptions

20. All project team members will participate throughout the four-month project period.
21. Team members will have the necessary basic programming and software engineering skills.
22. The required development tools and software will be available.
23. The pharmacy requirements will be available when needed.
24. The project supervisor will provide necessary academic guidance.
25. Team members will communicate regularly and attend project meetings.
26. The development team will have sufficient time to complete the planned features.
27. Users will provide feedback during requirements validation and system testing.
28. The project scope will remain reasonably stable after requirements approval.
29. Regular backups will be maintained during development and testing.
30. The system will be developed according to agreed requirements and within available resources.
31. Major requirement changes will be reviewed by the team leader before implementation.

# 13 Technical Requirements

| Area | Proposed Technology / Standard |
|---|---|
| **Frontend** | HTML/CSS/JavaScript and React.js |
| **Backend** | Django Framework |
| **Database** | MySQL |
| **Development** | Visual Studio Code |
| **Version Control** | Git / GitHub and later GitHub Actions |
| **Diagrams** | Draw.io / Lucidchart |
| **UI/UX** | Figma |
| **Authentication** | Secure login, password hashing and session/access control |
| **Authorization** | Role-based access control |
| **Security** | Input validation, secure database access and regular backups |

**ARCHITECTURE DECISION**

The exact technology stack can be finalized during the system-design phase after evaluating project requirements, team capability and deployment constraints.

# 14 Project Success Criteria

32. PMS is completed within the planned four-month period.
33. All essential requirements are implemented.
34. Users can successfully log in according to their roles.
35. Medicines can be added, updated, searched and managed.
36. Inventory can be accurately tracked.
37. Purchases and sales can be recorded.
38. Low-stock and expiration information can be identified.
39. Required reports can be generated.
40. The system passes functional and integration testing.
41. Critical bugs are resolved before final delivery.
42. The database maintains accurate and consistent information.
43. Technical and user documentation is completed.
44. The final system can be demonstrated successfully.
45. The project is delivered within the approved scope, schedule and available resources.

# 15 Summary Budget

| Project Component | Estimated Cost |
|---|---|
| **Project staffing** | Student project |
| **Development software** | 0 AFN |
| **Design and documentation tools** | 0 AFN |
| **Internet / communication** | About 2000 Afghani |
| **Hosting / deployment** | About 50$ |
| **Printing and documentation** | About 500 Afghani |
| **Miscellaneous** | Will be determined |

| Project Component | Estimated Cost |
|---|---|
| **TOTAL** | Will be finalized |

**BUDGET NOTE**

Most development software can be obtained through free/open-source or educational resources. Exact costs should be finalized after confirming hosting, printing, internet, hardware and other project expenses.

# 16 Communication Plan

| Communication | Frequency | Participants |
|---|---|---|
| **Team meeting** | Weekly | All team members |
| **Progress update** | Weekly | Team Leader + members |
| **Task assignment** | As required | Team Leader + members |
| **Requirements review** | During analysis | Team + stakeholders / supervisor |
| **Code review** | Regularly | Developers |
| **Testing review** | During testing | QA + developers |
| **Final review** | End of project | Entire team + supervisor |

The team may use WhatsApp, Telegram, Google Drive, GitHub or other agreed collaboration platforms.

# 17 Change Management

Any significant change to the approved project scope must be reviewed by the Team Leader / Project Manager.

- Reason for the requested change.
- Effect on project scope.
- Effect on project schedule.

- Effect on technical implementation.
- Effect on project quality.
- Effect on available resources.

Changes that significantly affect the four-month schedule should only be accepted after approval from the project supervisor / stakeholder.

# 18 Quality Management

Quality will be managed continuously rather than treated as a final-stage activity.

- Requirements review
- Code review
- Database validation
- Unit testing
- Integration testing
- System testing
- User acceptance testing
- Bug tracking
- Security testing
- Final system review

**RELEASE GATE**

All critical defects should be resolved before the final project demonstration and delivery.

# 19 Project Governance

## 19.1 Team Leader / Project Manager

Bahram Jafaree will be responsible for:

- Overall project coordination.
- Assigning tasks.
- Monitoring deadlines.
- Organizing meetings.
- Tracking project progress.
- Managing risks.
- Coordinating communication.
- Reviewing major deliverables.
- Ensuring the project remains within scope.

## 19.2 Development Team

The four team members will work collaboratively on analysis, design, development, testing and documentation according to their assigned responsibilities.

# 20 Project Approval

**APPROVAL STATEMENT**

By approving this Project Charter, the project stakeholders agree that the Pharmacy Management System (PMS) will be developed according to the defined objectives, scope, deliverables, schedule, assumptions and constraints.

## 20.1 Project Team Sign-off

| Name | Role | Signature |
|---|---|---|
| **Bahram Jafaree** | Team Leader / Project Manager | _____ |
| **Said Yousaf Hashimi** | System Analyst / Backend Developer | _____ |
| **Ali Seena** | Frontend / UI Developer | _____ |
| **Alireza Alizada** | Database Developer | _____ |
| **Zaki Ahmadi** | QA & Documentation | _____ |

## 20.2 Formal Approval

| Approval Item | Details |
|---|---|
| **Project Supervisor** | _____ |
| **Signature** | _____ |
| **Date** | _____ |

# 21 Project Charter Summary

| Dimension | Summary |
|---|---|
| **Project** | Pharmacy Management System (PMS) |
| **Duration** | 4 Months |
| **Team Leader** | Bahram Jafaree |
| **Team Members** | Said Yousaf Hashimi • Ali Seena • Alireza Alizada • Zaki Ahmadi |
| **Primary Goal** | Develop a secure, reliable and user-friendly system for medicines, inventory, purchases, sales, suppliers and reports. |
| **Major Deliverables** | Requirements, analysis, design, database, PMS software, testing, documentation and final demonstration. |
| **Primary Constraint** | Four-month completion deadline. |
| **Success Measure** | A tested and documented PMS that satisfies approved requirements and can be successfully demonstrated and used by pharmacy staff. |

## END OF PROJECT CHARTER

**Pharmacy Management System • Version 1.0 • Proposed**